# Capstone Project Monthly Progress Report

**Month:** February 2026

**Project Title:** Abnormal Crowd Behaviour Detection and Early Warning using Computer Vision

**Project ID:** CAPSTONE_2022_25196_1

**Student Name:** Manas Chowdary Kannikanti
**Student Roll Number:** AP22110010164
**Total Number of Students in Team:** 04
**Details of Team Members:**

A.Vaibhav (Roll No. AP22110010133)
P.Nagendra (Roll No. AP22110010527)
S.Srijan (Roll No. AP22110010131)

**Supervisor:** Dr.Surochita Pal

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## 1 Project Overview

The project focuses on developing a deep learning based system to detect abnormal crowd behaviour in dense public gatherings using surveillance video data. It aims to analyze motion patterns, density variations, and spatio-temporal features to identify unsafe crowd dynamics such as panic-like movement, congestion, and bottleneck formation. The system is designed to generate early warning signals before critical situations escalate.

## 2 Work Completed During Last 1 Month

- Finalized the project problem statement and research scope.
- Selected ShanghaiTech dataset for primary experimentation.
- Designed the overall system pipeline including motion analysis and risk scoring modules.
- Conducted literature review on crowd anomaly detection methods.
- Set up development environment and structured DGX training workflow.

## 3 Work Planned for this Month

- Implement video preprocessing and clip generation pipeline.
- Develop and train baseline anomaly detection model.
- Integrate optical flow for crowd motion feature extraction.
- Begin initial experimental evaluation on selected dataset.

## 4 Progress Status

- Overall completion: 60%
- Timeline structured and experimental phases planned.

## 5 Challenges/Issues Faced

- Identifying datasets that closely match large-scale dense crowd scenarios.
- Designing a meaningful risk scoring mechanism beyond simple anomaly detection.

## 6 Learning Outcomes

- Gained deeper understanding of video-based anomaly detection techniques.
- Learned about crowd motion modeling and density estimation concepts.
- Improved knowledge of research methodology and structured experimentation.

## 7 Plan for Next Month

- Complete baseline model training and evaluation.
- Implement risk scoring and early warning logic.
- Conduct ablation experiments.
- Prepare preliminary results for supervisor review.

## 8 Student Individual Contribution

Planned and structured the overall project framework, finalized dataset selection, and prepared the technical roadmap for implementation and experimentation.

## 9. Supervisor's Observation

**Student Completed:** ______________% of the planned/assigned task during the last month

**Comments:**

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**Number of times students meet with supervisor (Online or Offline):** _____

**Satisfied with the Student's Work:** Satisfied Not Satisfied

**Supervisor Signature:** ________________________

**Date:** DD/MM/YY